

YUANZHE HU

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EDUCATION

Georgia Institute of Technology

Ph.D. in Machine Learning

Advisor: Prof. Pan Li

Atlanta, GA

Aug. 2026 - Now

University of California, San Diego

M.S. in Computer Science and Engineering

Advisor: Prof. Zhiting Hu and Prof. Julian McAuley

La Jolla, CA

Sep. 2024 – Mar. 2026

Huazhong University of Science and Technology

B.S. in Artificial Intelligence (Qiming Honors College)

Wuhan, China

Sep. 2020 – Jun. 2024

RESEARCH SUMMARY

- **Long-term Memory for LLM Agents**^[1, 4, 6, 8]: Studied how LLM-agents memorize, update, and retrieve information over long horizons through systematic benchmarking and model building. Designed evaluation frameworks spanning single-session competency testing to multi-session interdependent agentic tasks, revealing that current systems generalize poorly in realistic settings. Built memory-augmented models scaling retention to 160K+ tokens and RL-trained agents generalizing to 400K+ tokens. Core contributor to MIRIX (3K+ GitHub stars), an open-source multi-agent memory framework.
- **Post-Training: SFT, RLVR, and RL for Agents**^[3, 8, 9]: Investigated post-training recipes for reasoning and agentic capabilities: built and analyzed large-scale SFT and RLVR pipelines (32B–70B, H200 clusters) achieving frontier reasoning performance; applied RL with verifiable rewards to train agent memory construction; designed layer-wise learning-rate balancing that improves low-resource fine-tuning by 2–10%.
- **Training Dynamics and Model Foundations**^[5, 7]: Analyzed failure modes and training dynamics in deep networks: identified three-regime patterns in SciML foundation models through loss-landscape geometry; proposed data-agnostic spectral weight signatures for model lineage tracking and performance quantification.

[·] refers to publications listed below.

PUBLICATION

* indicates equal contribution.

1. **Y. Hu***, Y. Wang*, J. McAuley “Evaluating Memory in LLM Agents via Incremental Multi-Turn Interactions” (ICLR 2026) [↗](#) [🔗 Agent Memory](#) [🔗 Agentic evaluation](#) [🔗 Long-context](#)
2. **Y. Hu**, K. Goel, V. Killiakov, Y. Yang “Eigenspectrum Analysis of Neural Networks without Aspect Ratio Bias” (ICML 2025) [↗](#) [🔗 LLM pruning](#) [🔗 Random Matrix Theory](#) [🔗 Training Dynamics](#)
3. Z. Liu*, **Y. Hu***, T. Pang, Y. Zhou, P. Ren, Y. Yang “Model Balancing Helps Low-data Training and Fine-tuning” (EMNLP 2024 Oral) [↗](#) [🔗 LLM fine-tuning](#) [🔗 Low-resource training](#)
4. Z. He*, Y. Wang*, C. Zhi*, **Y. Hu***, T.-P. Chen*, L. Yin, Z. Chen, T. A. Wu, S. Ouyang, Z. Wang, J. Pei, J. McAuley, Y. Choi, A. Pentland “Benchmarking Agent Memory in Interdependent Multi-Session Agentic Tasks” (ICML 2026) [🔗 Agent Memory](#) [🔗 Multi-turn Task](#) [🔗 Agent Evaluation](#)
5. Y. Wang*, **Y. Hu***, X. Zhong*, X. Wang*, H. Lu, T. Pang, M. W. Mahoney, Y. Yan, P. Ren, Y. Yang “Unveiling Multi-regime Patterns in SciML: Distinct Failure Modes and Regime-specific Optimization” (ICML 2026) [🔗 Scientific ML](#) [🔗 Training dynamics](#) [🔗 Loss landscape](#)
6. Y. Wang, D. Krotov, **Y. Hu**, Y. Gao, W. Zhou, J. McAuley, D. Gutfreund, R. Feris, Z. He “M+: Extending MemoryLLM with Scalable Long-Term Memory” (ICML 2025) [↗](#) [🔗 Memory-augmented LLM](#) [🔗 Long-term retention](#)

7. Z. Zhang*, I. V. Prasad*, **Y. Hu**, Z. Liu, H. Luo, P. Ren, Y. Yang “Spectral Signatures of Large Language Models” (**KDD 2026**)  [Model lineage](#)  [Spectral analysis](#)
8. Y. Wang, R. Takanobu, Z. Liang, Y. Mao, **Y. Hu**, J. McAuley, X. Wu “Mem- α : Learning Memory Construction via Reinforcement Learning” (**Preprint**)  [Agentic RL](#)  [Memory management](#)
9. LLM360 Team, Institute of Foundation Models, MBZUAI (incl. **Y. Hu**) “K2-Think: A Parameter-Efficient Reasoning System” (**Tech Report**)  [Large-scale reasoning](#)  [Post-training](#)

PROFESSIONAL EXPERIENCE

Institute of Foundation Models, MBZUAI

Remote

*Research Collaborator, supervised by **Hector Liu***

Jun. 2025 – Sep. 2025

- Engineered the large-scale SFT pipeline for 32B–70B reasoning LLMs (LLaMA / Qwen) on H200 clusters; led data pre-processing, model training, and technical report writing; analyzed training dynamics, data selection, and loss behavior (Paper: K2-Think).

UC San Diego – McAuley Lab

La Jolla, CA

*Research Assistant, supervised by **Yu Wang, Prof. Julian McAuley***

Oct. 2024 – Feb. 2026

- Studied long-term memory in LLM agents through systematic evaluation and model building: designed multi-turn and multi-session benchmarks, trained memory-augmented models scaling retention to 160K+ tokens, and applied RL to memory management achieving 52% accuracy on LongMemEval (S*) with one-third the context window. (Papers: MemoryAgentBench, M+, Mem- α , and MemoryArena)
- Core contributor to MIRIX (open-source, 3K+ GitHub stars), a multi-agent memory framework; responsible for memory evaluation pipeline design, benchmark integration, and metric development.

Dartmouth College

Hanover, NH

*Research Collaborator, supervised by **Prof. Yaoqing Yang** (with Dr. Pu Ren)*

Jun. 2023 – Apr. 2026

- Developed FARMS, an RMT-based subsampling method for unbiased, size-invariant layer-wise pruning, and a dynamic layer-wise learning-rate schedule for low-resource SFT; validated across LLM pruning, CV, and SciML (Papers: FARMS, Model Balancing, SciML Multi-Regime, and LLM Signatures).

SERVICES AND AWARD

Reviewers: TMLR, ICLR/NeurIPs/ICML/KDD 2026, ACL Rolling Review 2025/2026, and Workshops on ICML/NeurIPs/ICLR.

Award: 2nd Prize (Top 7% among 2,000+ teams), 17th National College Students Intelligent Car Race, Awarded by Chinese Automation Association, 2022 — *Team Leader*

SKILLS

Programming Languages: Python, C/C++, SQL, Bash, MATLAB

ML Tools & Frameworks: PyTorch, Transformers, LLaMA-Factory, VeRL, DeepSpeed, FSDP, vLLM, SGLang